THE PHYSICS OF FLUIDS VOLUME 2, NUMBER 2 MARCH-APRIL, 1959

Blast Wave from a Spherical Charge

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(Received July 10, 1958; revised manuscript received December 10, 1958)

The blast wave from the detonation of a spherical charge of TNT is described based on results of a numerical calculation. The equations of motion and the equations of state for TNT and for air are described. The pressures, densities, temperatures, and velocities are detailed as functions of time and radius. Space-time relations and energy and impulse histories are shown. A second shock is seen to originate as an imploding shock following the inward rarefaction into the explosion product gases and a series of subsequent minor shocks are seen to appear in a similar manner, moving out in the negative phase behind the main shock.